

A generic detachable attachment for existing wheelchairs

to enable moving up staircases



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E/21/084
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The Problem

- Standard mobility solutions fail to provide affordable stair-climbing capabilities, forcing users to rely on manual lifting that poses significant safety risks and physical strain for both the user and the helper.

Our Aim

- **Universal Fit:** Quick-clamp compatibility for all manual chairs.
- **Lightweight:** <10kg assembly using high-strength alloys.
- **User Comfort:** Ergonomic, stable, and secure Horizontal seating experience.
- **Fail-Safe:** Integrated automatic incline braking system.
- **Motorized:** Power-assisted mechanism for effortless helper operation.
- **Agile:** Designed for tight U and L-turn navigation.
- **Low Cost:** Optimized for high manufacturing and market viability.

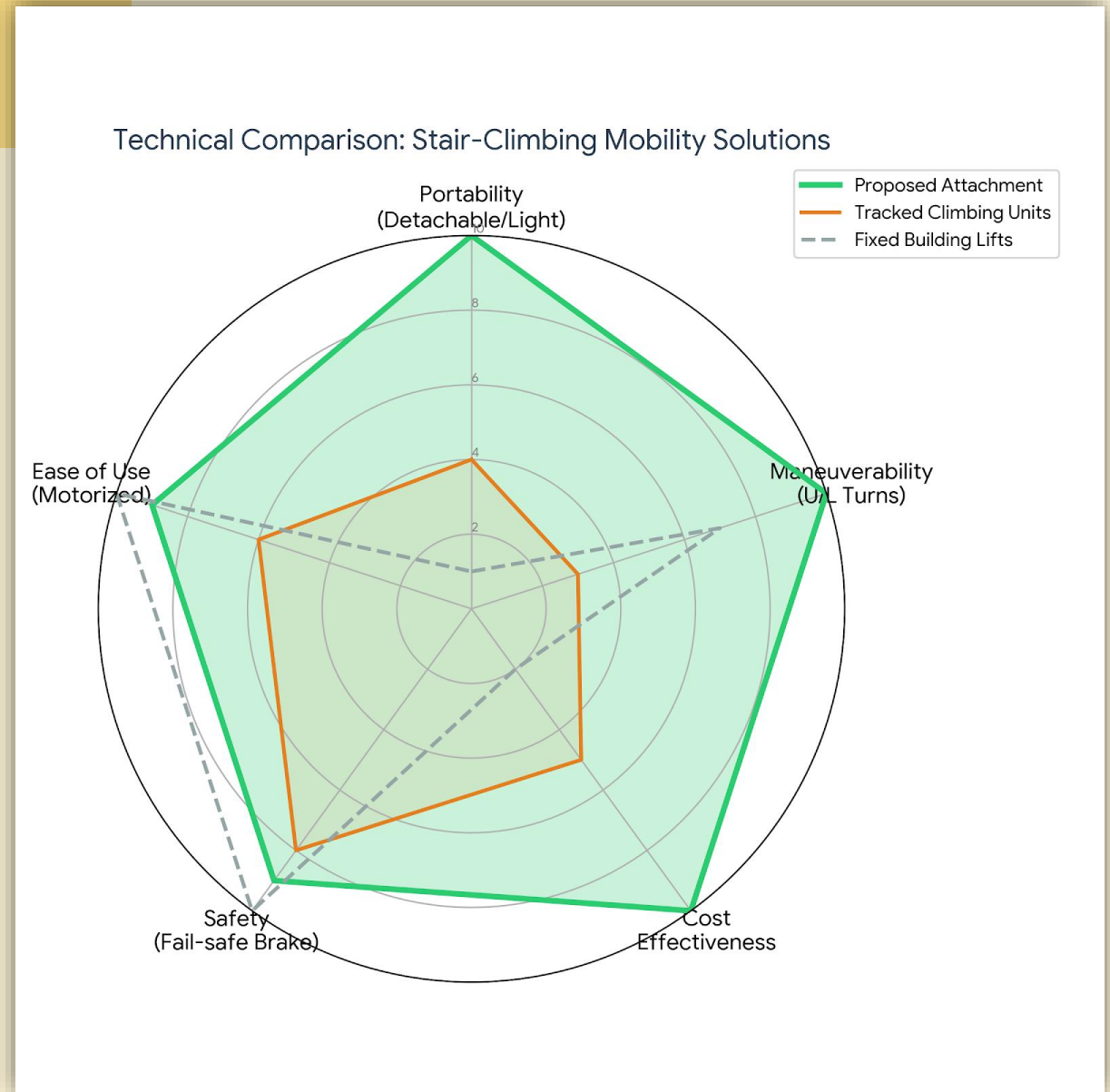


Figure 1: Performance Benchmarking: Proposed Attachment vs. Existing Mobility Solutions

Engineering Design & Methodology



Design and Simulation

- Stair-Climbing Mechanism & Linkage Design
- Universal Quick-Clamp Frame Architecture
- Kinematic and Path-Mate Motion Simulation
- SolidWorks 3D CAD Modeling

Implementation

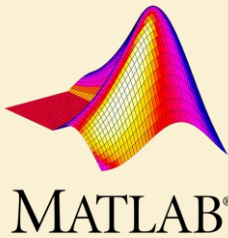
- Fabrication of High-Strength Alloy Frame
- Control System Programming (Arduino/Micro controller)
- Stair-Drive Motor & Gearbox Unit Integration

Testing

- Bench Testing (No-load Motor & Drive Logic)
- Static Load Testing (Structural Integrity Verification)
- Dynamic Stair-Climbing Tests with Variable Loads
- Fail-Safe Brake Response

Evaluation

- Horizontal Stability Accuracy Analysis
- Battery Efficiency & Power Consumption per Climb
- Turn Radius & Maneuverability Evaluation (U/L Turns)
- Performance Comparison with Manual Stair-Ascent Methods



Timeline

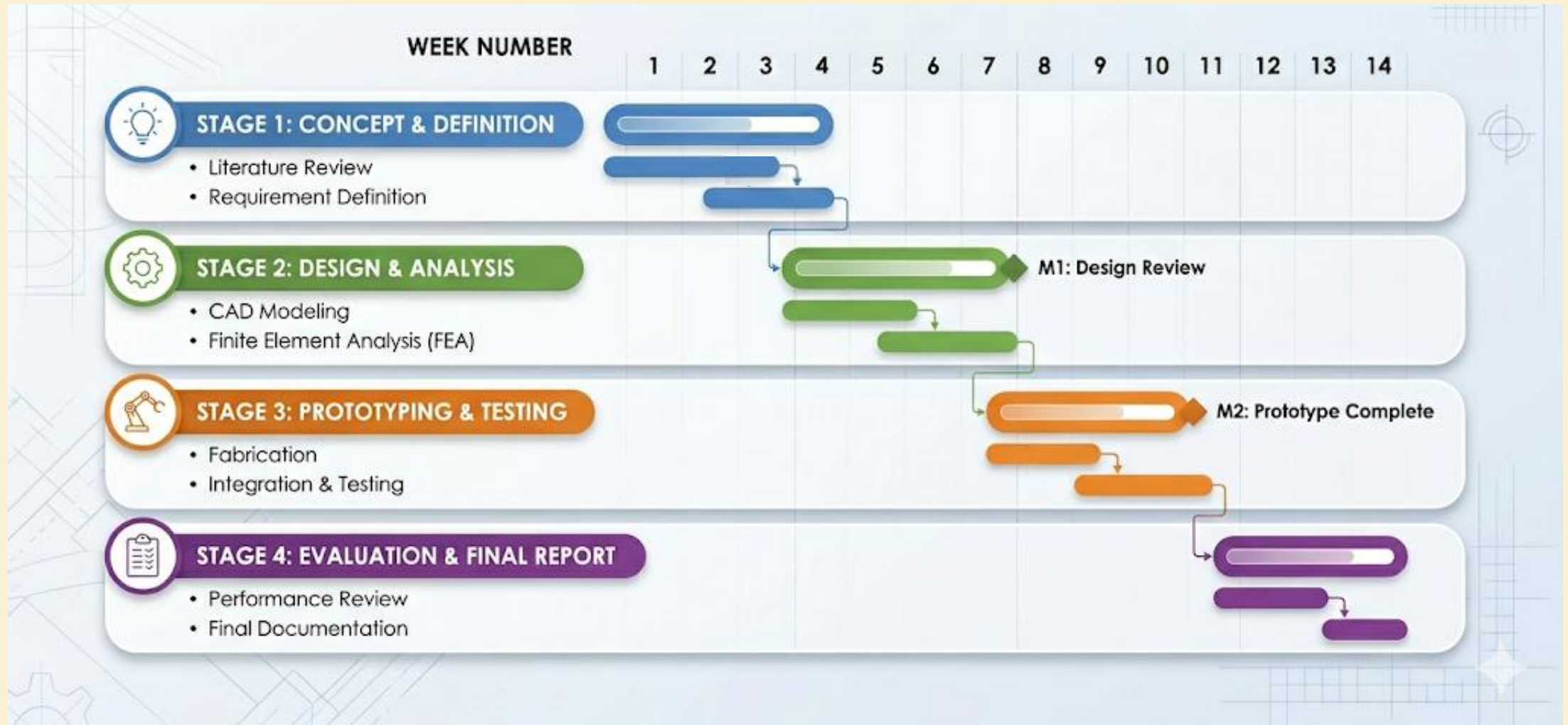


Figure 2: Gantt bar of Timeline of the project

ECONOMIC



- **Manufacturing Feasibility:** Designed for low-cost production using standard components to ensure market viability.
- **Cost-Efficiency:** Provides an affordable alternative to permanent lifts or integrated power chairs.



SOCIAL



- **End-User Safety:** Features a fail-safe braking system and ergonomic attachment points for maximum user comfort.
- **Ease of Use:** Motorized power mechanism reduces helper fatigue and minimizes physical strain.



ENVIRONMENTAL



- **Sustainability:** Utilizes lightweight, recyclable alloys to reduce material waste during fabrication.
- **Energy Efficiency:** Optimized power-train designed for minimal battery consumption and reduced carbon footprint.



Figure 3: The "Big Three" Impact

Q & A